

Abstract Machine : TOC.

- > ① F.A. Finite Automaton $\begin{cases} \text{D.F.A.} \\ \text{N.F.A.} \end{cases}$
- > ② P.D.A Pushdown Automaton $\begin{cases} \rightarrow \text{D.P.D.A} \\ \rightarrow \text{NPPA} \end{cases}$
- ③ Turing Machine.

F.A.
State transition Fn
Has
no
memory

P.D.A.
State transition funct.
 stack
memory

Turing.

S.T.F
+ $\equiv$

R.A.M.

Turing
Machine.
1920's.

Alonzo Church
1930

λ -calculus

1936

Turing $\equiv$ λ -calculus

CHURCH-TURING EQUIVALENCE

F.A. $\rightarrow$ P.D.A. $\rightarrow$ T.M.

State Change.

Memory

Instructions

Memory State Read. / Write:

load / store

mov

$\% =$ $=$

$\boxed{=}$ $==$

F.A. $\rightarrow$ D.F.A.
 $\rightarrow$ N.F.A

P.O.A. $\rightarrow$ D.P.D.A.
 $\rightarrow$ N.P.O.A.

T.M.

load / store.

mov.

$\% =$

$=$

λ -calculus

functional
program

λ -expression

V.M.

R.M. $\odot$

H/W Scaling.

thread.

Shared
data
update

HPC